

Portfolio Project

Microsoft Excel | Data Cleaning | Data Validation | Business Analysis |
Dashboard Development

COSSM- ETICS

Sales Data Cleaning & Analysis

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• Project Summary:

Cosmetics Sales is a beauty retail business operating through both online and offline sales channels. The sales dataset was collected from multiple sources, resulting in inconsistent formatting, missing values, and data quality issues that affected reporting accuracy.

The objective of this project was to clean, standardize, validate, and prepare the dataset for reliable business analysis using Microsoft Excel. The project also included calculating key business metrics and building an interactive dashboard.

The final deliverable is a clean, analysis-ready dataset supported by accurate business insights, enabling more reliable reporting and better business decision-making.

• Business Background:

The company operates as a beauty retailer offering cosmetic products through both physical stores and online sales channels. As customer orders increased, sales information was gathered from multiple systems, creating inconsistencies across the dataset.

Because the data originated from different sources, several quality issues appeared, including inconsistent date formats, missing values, incorrect data types, invalid quantities, and inconsistent text formatting. These problems made the dataset unreliable for reporting and business analysis.

To support accurate decision-making, the business required a structured data cleaning process that would improve data quality, standardize information, and prepare the dataset for meaningful business reporting

• Business Problem:

1. The original dataset contained:
2. Missing values
3. Mixed date formats
4. Invalid quantities
5. Incorrect price formats
6. Missing product information

These issues reduced reporting accuracy and required a structured data cleaning process before analysis.

• Client Requirements:

The client requested a complete data cleaning and business analysis solution to improve data quality and prepare the sales dataset for reliable reporting. The project scope included data cleaning, validation, enrichment, and the calculation of key business metrics to support informed decision-making.

We are a growing beauty retail business operating both online & offline. We have recently collected sales data from multiple sources but the dataset contains inconsistencies, missing values, and formatting issues.

We need a Data Analyst (freelancer) to clean, standardize, and analyze the dataset to make it suitable for business reporting and decision-making.

FIGURE 1. ORIGINAL CLIENT PROJECT BRIEF

Data Cleaning: • Standardize all date formats into YYYY-MM-DD. • Convert price values into numeric format. • Handle missing values appropriately. • Standardize text formatting across all categorical fields. • Remove or correct invalid quantities and inconsistent records.	Data Validation: • Identify rows containing missing values. • Detect inconsistent or logically incorrect records. • Highlight incomplete product information. • Verify data consistency after cleaning.
Data Enrichment: • Infer missing product sizes where possible. • Complete missing values using logical assumptions. • Label unresolved values as "Unknown".	Business Analysis: • Calculate Total Sales. • Identify Top-Selling Products. • Determine Top Customers. • Analyze Sales Distribution by Category.

• Dataset Overview:

Before performing any data cleaning or analysis, the dataset was reviewed to understand its structure, identify key fields, and assess the overall data quality. The dataset contains order-level sales records collected from multiple sources, making it necessary to evaluate each column before applying cleaning and validation techniques.

Dataset Information	Details
Industry	Beauty Retail
Dataset Type	Sales Transactions
Data Source	Multiple Sales Sources
Number of Columns	9
Primary Purpose	Data Cleaning & Business Analysis

• Key Observation:

The initial review showed that the dataset contained multiple data quality issues, including missing values, inconsistent formatting, incorrect data types, and invalid records. These observations defined the scope of the data cleaning and validation process.

• **Data Quality Assessment:**

Before applying any cleaning techniques, the dataset was assessed to identify data quality issues that could affect reporting accuracy and business analysis. The assessment focused on detecting missing values, inconsistent formatting, invalid records, incorrect data types, and incomplete product information.

Issue	Description
Inconsistent Date Formats	Multiple date formats were used across the dataset.
Missing Values	Several important fields contained blank values.
Invalid Data Types	Some price values were stored as text instead of numbers.
Negative Quantities	Invalid negative quantities were identified.
Inconsistent Text Formatting	Differences in capitalization and text formatting existed across categorical fields.
Incomplete Product Information	Some product attributes, such as size, were missing or incomplete.

• **Assessment Summary:**

The assessment confirmed that the dataset required comprehensive cleaning and standardization before it could be used for reporting or business analysis. These findings formed the basis of the data cleaning strategy implemented throughout the project

Sheet 1: Orders_Raw							
City	Customer	Quantity	Price	Size	Category	Product Name	Order ID
cairo	Sara Ahmed	2	120	Small	Makeup	Lip Gloss	2001
Cairo	Sara Ahmed	2	120	Small	Makeup	Lip Gloss	2002
Giza	Mona Ali	1	250	50ml	Skincare	Face Cream Rose	2003
Cairo	Omar Khaled	1	"two hundred"	100ml	Haircare	Hair Oil	2004
Cairo	Sara Ahmed	1-	300	200ml	Body Care	Body Lotion Ocean	2005
Cairo		2	180	100ml	Skincare	Face Mask Clay	2006
Alex	Nour Hassan	1	550	50ml	Fragrance	Perfume Vanilla	2007
Giza	Mai Adel	1	150	Small	Makeup	Lipstick Red	2008
cairo	Omar Khaled	2	220	75ml	Haircare	Hair Serum	2009
Alexandria	Youssef Ali	1	280	250ml	Body Care	Body Scrub Coffee	2010

FIGURE 2. EXAMPLES OF DATA QUALITY ISSUES IN THE RAW DATASET

• Data Cleaning Process:

After identifying the data quality issues, a structured cleaning process was applied to improve accuracy, consistency, and usability. The dataset was standardized and prepared for reliable business analysis.

-Cleaning Techniques Performed:

1. Standardized all date formats.
2. Converted price values to numeric format.
3. Handled missing values using logical assumptions where possible
4. Corrected invalid quantities.
5. Standardized text formatting across categories, cities, and product names.
6. Marked unresolved values as “Unknown” when necessary.

-Result:

The cleaning process produced a consistent and analysis-ready dataset that could be used confidently for reporting, dashboard creation, and business decision-making

Order ID	Product name	Category	Size	Price	Quantity	Customer Name	City	Order Date	Transaction Type	Sales Amount
2002	Lip Gloss	Makeup	Small	120	2	Sara Ahmed	Cairo	01/02/2026	Sale	240
2003	Face Cream Rose	Skincare	50ml	250	1	Mona Ali	Giza	02/02/2026	Sale	250
2004	Hair Oil	Haircare	100ml	200	1	Omar Khaled	Cairo	03/02/2026	Sale	200
2005	Body Lotion Ocean	Body Care	200ml	300	-1	Sara Ahmed	Cairo	03/02/2026	Return	-300
2006	Face Mask Clay	Skincare	100ml	180	2	unknown	Cairo	04/02/2026	Sale	360
2007	Perfume Vanilla	Fragrance	50ml	550	1	Nour Hassan	Alexandria	02/05/2026	Sale	550
2008	Lipstick Red	Makeup	Small	150	1	Mai Adel	Giza	05/02/2026	Sale	150
2009	Hair Serum	Haircare	75ml	220	2	Omar Khaled	Cairo	06/02/2026	Sale	440
2010	Body Scrub Coffee	Body Care	250ml	280	1	Youssef Ali	Alexandria	07/02/2026	Sale	280

FIGURE 3. CLEANED AND STANDARDIZED DATASET

• Data Validation:

After the data cleaning process was completed, the dataset was reviewed to verify that all modifications were applied correctly. Validation checks ensured that the data was complete, consistent, and suitable for business reporting and analysis.

Validation Check	Result
Missing values reviewed	✓
Date formats standardized	✓
Price values converted to numeric	✓
Invalid quantities corrected	✓
Text formatting standardized	✓
Unknown values clearly labeled	✓

• Validation Outcome:

The validation process confirmed that the dataset was consistent, reliable, and ready for business analysis. All identified data quality issues were successfully addressed according to the project requirements.

• Business Analysis & Dashboard:

After cleaning and validating the dataset, key business metrics were calculated to provide meaningful insights into sales performance. An interactive Excel dashboard was created to present the results in a clear and easy-to-understand format.

**Total Sales:**
Calculated total revenue using Price × Quantity for each transaction.
“84,600 EGP”

**Top-Selling Products:**
Identified the products with the highest sales performance.
“Perfume Vanilla”

**Top Customers:**
Ranked customers based on their total spending.

**Sales by Category:**
Analyzed sales distribution across different product categories.
“Haircare”

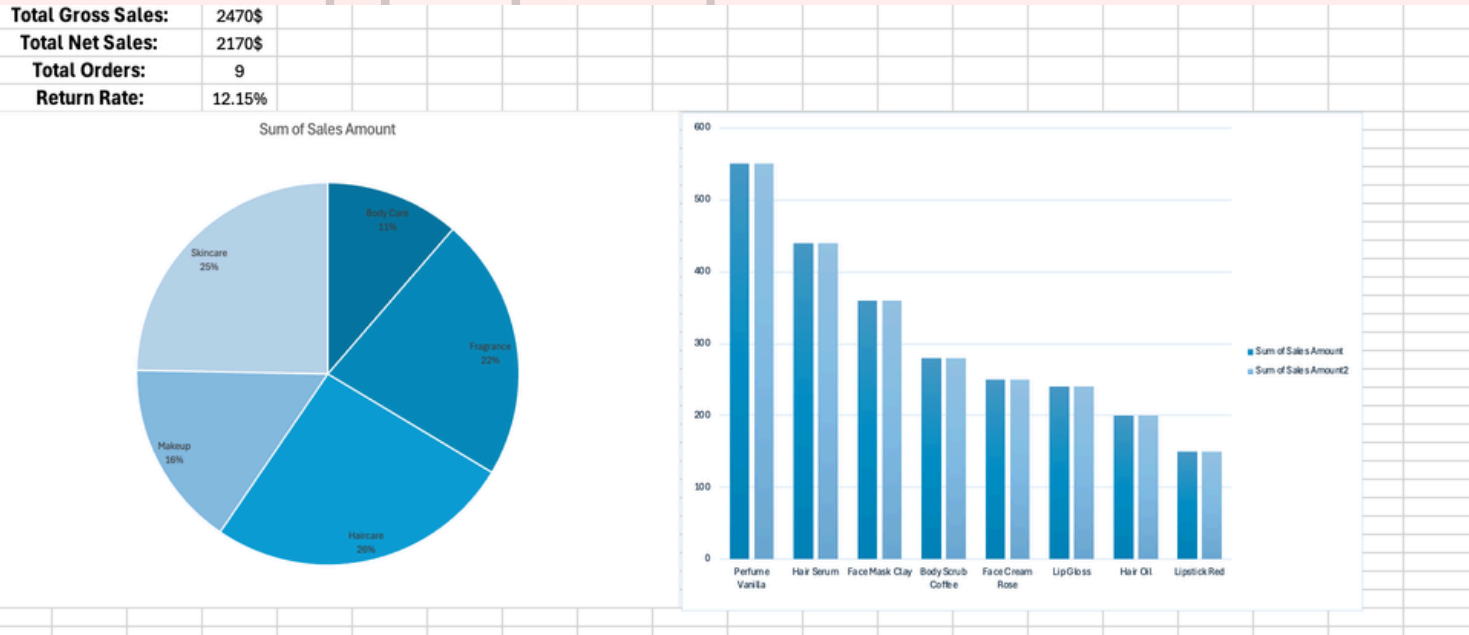


FIGURE 4. INTERACTIVE EXCEL SALES DASHBOARD

• Key Insights:

The dashboard transformed the cleaned dataset into meaningful business insights, enabling quick performance monitoring and supporting data-driven decision-making through clear visual reporting.

• Business Impact:

The completed data cleaning and analysis process significantly improved the quality, consistency, and reliability of the sales dataset. By transforming raw and inconsistent data into a structured and analysis-ready format, the business can now generate accurate reports and make more informed decisions based on reliable information.

The cleaned dataset can now be used confidently for future reporting, sales monitoring, and performance analysis without additional manual corrections.

 Improved Data Quality: Standardized and validated data reduced inconsistencies and improved overall accuracy.	 Reliable Business Reporting: Clean data enabled accurate sales calculations and meaningful performance analysis.
 Better Decision-Making: Clear business insights support faster and more confident decision-making.	 Organized Data Structure: A consistent dataset makes future updates, reporting, and maintenance much easier.

• Project Outcome:

The project successfully transformed a raw sales dataset into a clean, standardized, and business-ready solution. The final Excel workbook provides reliable data, meaningful insights, and an interactive dashboard that supports ongoing business reporting and operational efficiency.

• Tools & Skills Used:

-Tools:

1. Microsoft Excel
2. Pivot tables
3. Charts
4. Conditional Formatting
5. Excel Functions
6. Data Validation
7. Data Analysis
8. Dashboard Development

-Skills:

1. Data Cleaning
 2. Data Standardization
 3. Data Validation
 4. Business Analysis
 5. Dashboard Design
 6. Data Organization
 7. Problem Solving
 8. Attention to Detail
 9. Critical Thinking
 10. Logical Problem Solving
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This project demonstrates practical experience in transforming raw business data into a structured, analysis-ready dataset using Microsoft Excel. It highlights the ability to clean, validate, organize, and analyze data while creating meaningful business insights through professional reporting and dashboard development.

• Conclusion:

This project demonstrates a complete data cleaning and business analysis workflow using Microsoft Excel. Starting from a raw and inconsistent dataset, the project focused on improving data quality, validating business records, and transforming the data into a reliable source for reporting and analysis.

By applying structured data cleaning techniques and creating an interactive dashboard, the final solution provides accurate business insights that support better decision-making and improve reporting efficiency. The project reflects practical analytical skills that can be applied to real-world business scenarios.

This project reflects my ability to transform raw business data into reliable business insights through structured analysis and professional Excel reporting.

• Key Achievements:

- ✓ Cleaned and standardized a raw sales dataset.
- ✓ Validated and enriched business data.
- ✓ Calculated key sales metrics.
- ✓ Designed an interactive Excel dashboard.
- ✓ Delivered a professional reporting solution.



THANK YOU

Thank you for reviewing this project.

This portfolio project demonstrates my practical skills in Data Cleaning, Data Validation, Business Analysis, and Dashboard Development using Microsoft Excel.

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